

Intellectual Traditions in Ancient Greek and Latin texts: A Co-occurrence Network Approach

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Introduction

The study of (Social) Networks has steadily gained traction in the study of Graeco-Roman antiquity. For instance, the spread of religions or cults throughout antiquity has been studied with approaches from network theory based on epigraphic attestations in certain places of certain deities of these religions and cults (Collar 2013; 2022). Other studies in the field of prosopography employ quantitative network approaches to reconstruct Athenian kinship networks determined from naming traditions (Karila-Cohen 2018), to facilitate the identification of individuals in cases of homonymy, or to trace the spread of certain personal names in Greco-Roman Egypt (Broux and Depauw 2015; Broux 2019). In the study of literary texts, the focus is predominantly on the manual reconstruction of networks of people based on real-life interactions or relations with traditional close reading methods (Broekaert, Köstner, and Rollinger 2020; Guérin 2024; McClure 2020; Rollinger 2014). However, even in the study of literary texts, automatically constructed networks are emerging as well, exploring other types of connections in these texts (Beine 2024; Dörpinghaus 2022).

In this study, I model a co-occurrence network of individuals in Ancient Greek and Latin texts, similar to Dörpinghaus' (2022) approach for biblical narratives. This functions as a case study within the NIKAW (Networks of Ideas and Knowledge in the Ancient World) project that aims to study the flow of knowledge in Ancient Greek and Latin literature using Natural Language Processing (NLP) and Social Network Analysis (SNA) methods. This case study serves as proof of concept for the applicability of the co-occurrence network model, addressing concerns about the interpretability and meaningfulness of networks with edges based on textual proximity (Dörpinghaus 2022, 8; Guérin 2024, 8). As I aim to assess the usefulness of co-occurrence networks in understanding the reception of pivotal authors over time, Plato, a prolific author with well-documented impact, serves as a model test case. Additionally, his influence on both Christian and non-Christian authors allows for a comparative analysis of co-occurrences within two distinct intellectual traditions

A short paper based on this work has also been submitted to the DH2025 conference. Whereas that paper will focus on the comparison of Plato's co-occurrence networks before and after the advent of Christianity, this longer paper will focus on overarching methodological considerations. Specifically, it aims to lay the theoretical and methodological groundwork for any further applications of such co-occurrence networks.

Data

The corpus, detailed in Table 1 below, was constructed to meet several criteria: texts must contain the lemma 'Πλάτων' / 'Platon' more than five times, span a wide diachronic range, vary in text type (limited to prose) and authored both Christians and non-Christians. Individuals in these texts are manually annotated and disambiguated by linking them to the Wikisource edition of *Paulys Realencyclopädie der classischen Altertumswissenschaft*. For shorter texts, all individuals were annotated, while for longer texts, only those occurring in sub-sections that mention Plato.

Author	Work	Period	Christian work
Greek			
[Plato]	<i>Epistulae</i>	BCE	☐
Aristoteles	<i>Metaphysica</i>	BCE	☐
Strabo	<i>Geographica</i>	BCE	☐
Dionysius Halicarnassensis	<i>De Demosthenis dictione</i>	BCE	☐
Plutarchus	<i>Quaestiones convivales</i>	CE	☐
Claudius Aelianus	<i>Varia historia</i>	CE	☐
Clemens Alexandrinus	<i>Stromata</i>	CE	☒
Origenes	<i>Contra Celsum</i>	CE	☒
Diogenes Laertius	<i>Vitae philosophorum</i>	CE	☐
Galenus	<i>De placitis Hippocratis et Platonis</i>	CE	☐
Latin			
Cicero	<i>Tusculanae Disputationes</i>	BCE	☐
Tertullian	<i>De Anima</i>	CE	☒
Seneca	<i>Ad Lucilium Epistulae Morales</i>	CE	☐
Lactantius	<i>Divinarum Institutionum</i>	CE	☒
Apuleius	<i>Pro Se De Magia Liber</i>	CE	☐

Table 1: Corpus for co-occurrences with Plato.

This annotation facilitates the construction of a multi-modal graph as represented in the schema displayed in Figure 1. The nodes in this graph, represented by the circles, are Persons, Works, and Mentions and listed together with associated attributes. The attributes listed in this figure are still in the process of being collected systematically and can at the time of writing thus not yet be explored fully. The edges in the original multi-modal graph encode the following relationships:

- A work contains a mention of a personal name: (:Work) – [:CONTAINS] -> (:Mention)
- The mention refers to an individual: (:Mention) – [:REFERS_TO] -> (:Person)
- A person wrote a work: (:Person) – [:WROTE] -> (:Work)

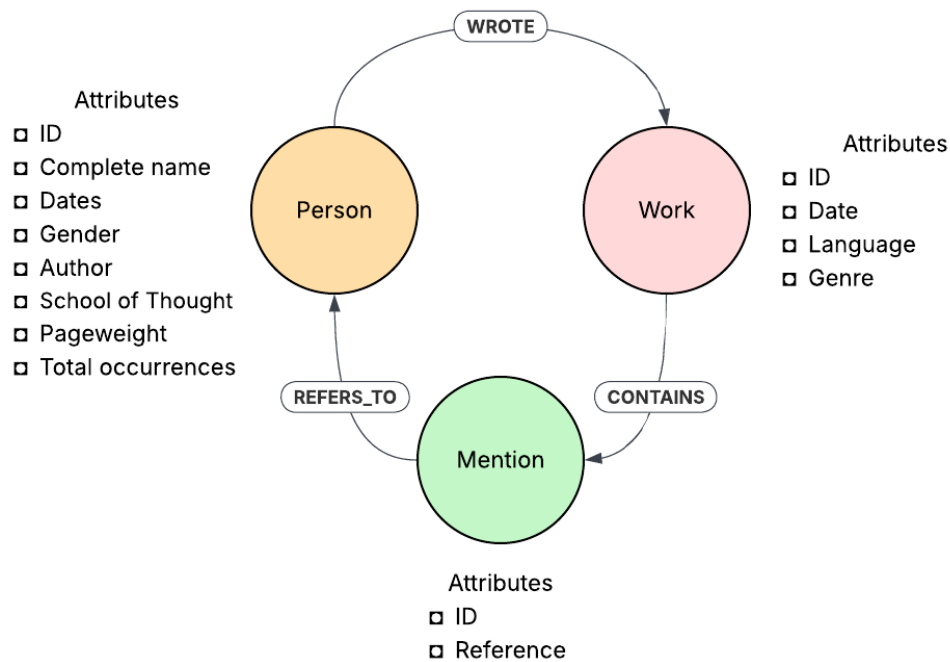


Figure 1: Schema of the multi-modal graph for mentions of people.

From this multi-modal graph we can project co-occurrence networks of persons at various levels of textual proximity. This projection would contain only Person nodes with a CO_OCCURS_WITH relation between two distinct individuals if they co-occur within a specific section of a text. This paper will focus specifically on a preliminary description and analysis of this Person-Person co-occurrence projection. Edges are based on co-occurrence between individuals in sections of the text that mention Plato explicitly, thus limiting the network to individuals that occur in similar contexts as Plato.

Methodology

The primary objective of this paper is to explore the formal aspects of co-occurrence networks and address the interpretability of various levels of textual proximity. For this, I will compare the co-occurrence networks to co-citation networks, commonly studied with SNA approaches in the field of bibliometrics. In co-citation studies, Author Co-Citation Analysis (ACA) can be performed with SNA methods to uncover intellectual structures within modern academic fields (Kim and Barnett 2008; White and McCain 1998). For example, Kim and Barnett (2008) analyze a co-citation network to conclude that detected clusters line up with previous findings that the field of Communication studies consists of two distinct groups, one focusing on mass media and one on interpersonal communication. More recently, the Citation Proximity Analysis (CPA) approach has suggested that co-citation within smaller units of texts such as sections, paragraphs, and sentences, indicates greater similarity and closer relations between the two cited entities (Gipp and Beel 2009; Kim, Jeong, and Song 2016; Thijs 2020). These methods of the field of bibliometrics aim to identify (sub-)disciplines and track changes within these fields.

This paper presents an initial assessment of whether networks constructed using the ACA and CPA methodologies can assist in identifying specific intellectual traditions or clusters. Preliminary experiments already demonstrate that a large-scale evaluation to determine whether Plato's co-occurrence with certain individuals reflects distinct aspects of his philosophy or its reception, requires systematically recorded attributes. To enable this for future experiments, I am employing a semi-automated approach to attribute collection, leveraging connections to external resources like the domain specific [Trismegistos](#) (TM) database and more general resources such as Wikidata and Wikipedia. In the presentation, I show what can already be achieved by analyzing one attribute native to our dataset and another derived from Trismegistos for the Greek part of the corpus. First, I will examine individuals who appear exclusively in Christian texts, exclusively in non-Christian texts, and in both, to explore how these categories align with modularity-based clustering. Second, I will analyze author positions in the network using the [TM Author ID](#). These preliminary investigations will illustrate how such attributes manifest within the network structure.

Conclusion

This paper draws on a case study of co-occurrence patterns surrounding Plato to highlight both the potential and the limitations of this network-based approach. The experiments presented demonstrate how analyzing attributes can shed light on the origins of structural patterns within the network. Looking ahead, I plan to expand these preliminary investigations to assess whether co-occurrence networks can reveal meaningful intellectual structures, similarly to how ACA and CPA uncover (sub-)disciplines in contemporary academic contexts.

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